



(Somaiya Vidyavihar University)

Batch: A2 Roll No.: 16010324044

Experiment No :- 6

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Sampling (in time and frequency) and analysis of Aliasing.

AIM: Analysis of the spectrum of the Sampled signals, effects of aliasing, effect of shift of samples in time, on magnitude of frequency spectrum. Finding a need of low pass filter before sampling / band limited signals for sampling

OUTCOME: Analyse discrete time systems in time and Z-domain.

PROCEDURE:

1. Sampling: (hint- replacing continuous index t of continuous function $x(t)$ with nT_s)
2. Sampling and high frequency noise; (hint- An alias of high frequency noise will show up in
3. sampled signal)
4. Aliasing : (hint- $f + k f_s$, k -integer),
5. Nyquist rate: (hint – $f_s \geq 2(\text{bandwidth})$ maximum frequency of signal)
6. Band pass sampling (hint –limited bandwidth).

Code:

```
clc;
clear;
close all;
%% 1. Continuous-Time Signal and Sampling
f = 10;                      % Signal frequency = 10 Hz
t = 0:0.001:1;              % Continuous-time axis
x = sin(2*pi*f*t);
% Sampling
fs = 50;                      % Sampling frequency = 50 Hz
Ts = 1/fs;
n = 0:fs;                      % Sample indices
tn = n*Ts;                      % Sampling instants
xn = sin(2*pi*f*tn);
% Plot
figure;
plot(t, x, 'b', 'LineWidth', 1);
hold on;
stem(tn, xn, 'r', 'filled');
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Sampling of a Continuous-Time Signal');
legend('Continuous signal x(t)', 'Sampled signal x(nT_s)');
%% 2. High-Frequency Noise and Aliasing
fn = 100;                      % Noise frequency = 100 Hz
noise = 0.5*sin(2*pi*fn*t);
x_noisy = x + noise;
```

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% Sample the noisy signal
xn_noisy = sin(2*pi*f*tn) + 0.5*sin(2*pi*fn*tn);
% Plot
figure;
subplot(2,1,1);
plot(t, x_noisy, 'b', 'LineWidth', 1);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal with High-Frequency Noise');
subplot(2,1,2);
stem(tn, xn_noisy, 'r', 'filled');
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Sampled Signal - High-Frequency Noise Aliasing');
%% 3. Aliasing Demonstration
fs1 = 100;           % Sampling frequency = 100 Hz
Ts1 = 1/fs1;
f11 = 20;           % Original frequency = 20 Hz
k = 1;
f21 = f11 + k*fs1;   % Aliased frequency = 120 Hz
% Continuous-time axis
t1 = 0:0.0001:0.2;
x11 = sin(2*pi*f11*t1);
x21 = sin(2*pi*f21*t1);
% Sampling instants
n1 = 0:20;
tn1 = n1*Ts1;
% Sampled signals
x1n = sin(2*pi*f11*tn1);
x2n = sin(2*pi*f21*tn1);
% Plot
figure;
plot(t1, x11, 'b', 'LineWidth', 1.2);
hold on;
plot(t1, x21, 'r--', 'LineWidth', 1.2);
stem(tn1, x1n, 'ko', 'filled');
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Aliasing: f and f + kf_s');
legend('20 Hz', '120 Hz', 'Samples');
%% 4. Band-Pass Sampling
fL = 80;           % Lower frequency = 80 Hz
fH = 100;          % Upper frequency = 100 Hz
fc = (fL + fH)/2;   % Center frequency
B = fH - fL;        % Bandwidth
% Sampling frequency
fsb = 60;
Tsb = 1/fsb;
% Continuous-time signal
tb = 0:0.0001:0.2;
% Band-pass signal
xb = cos(2*pi*fL*tb) + cos(2*pi*fH*tb);
% Sampling instants
nb = 0:floor(0.2/Tsb);
tnb = nb*Tsb;
% Sampled band-pass signal
xnb = cos(2*pi*fL*tnb) + cos(2*pi*fH*tnb);
% Plot
figure;

```

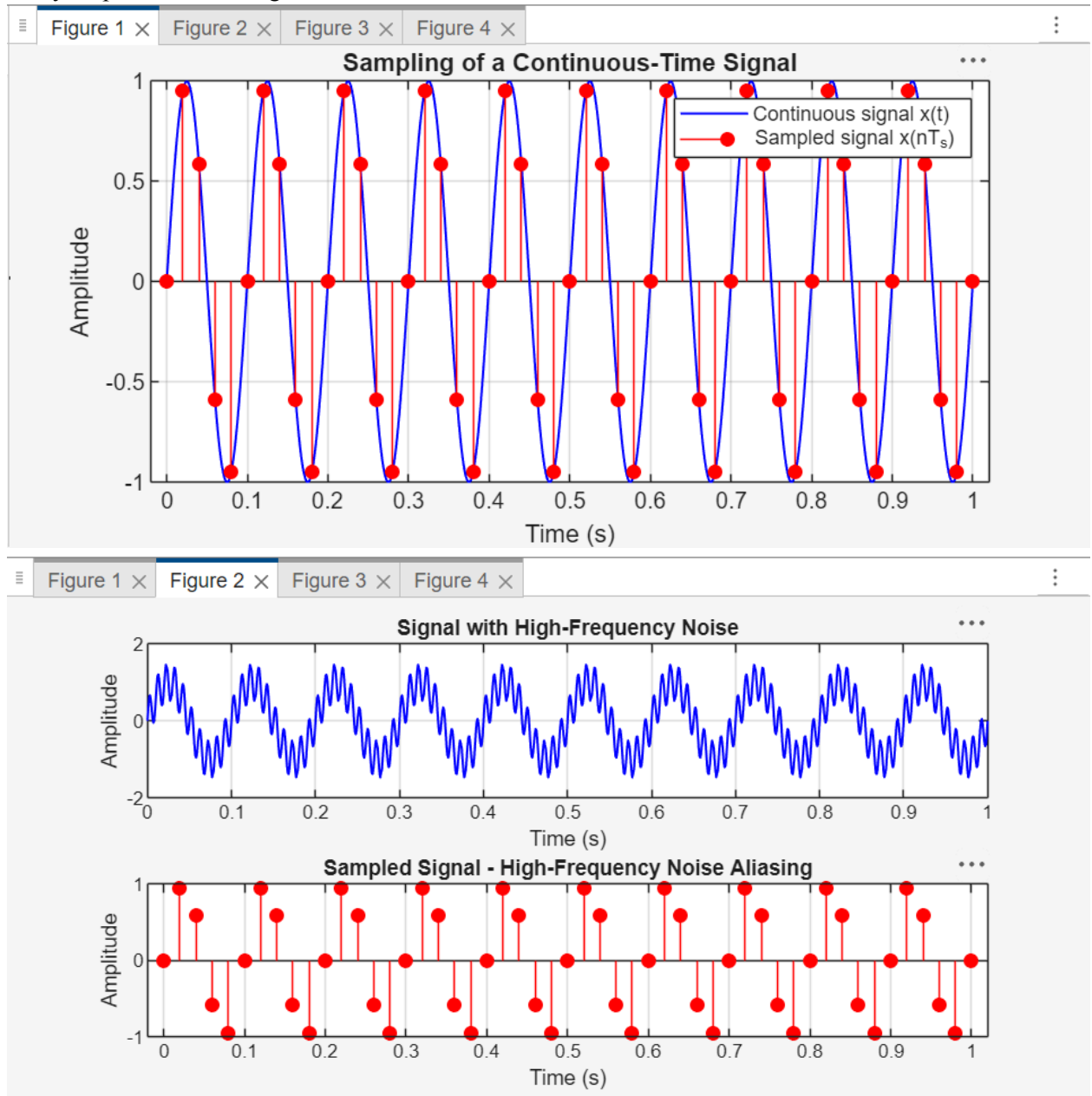
```

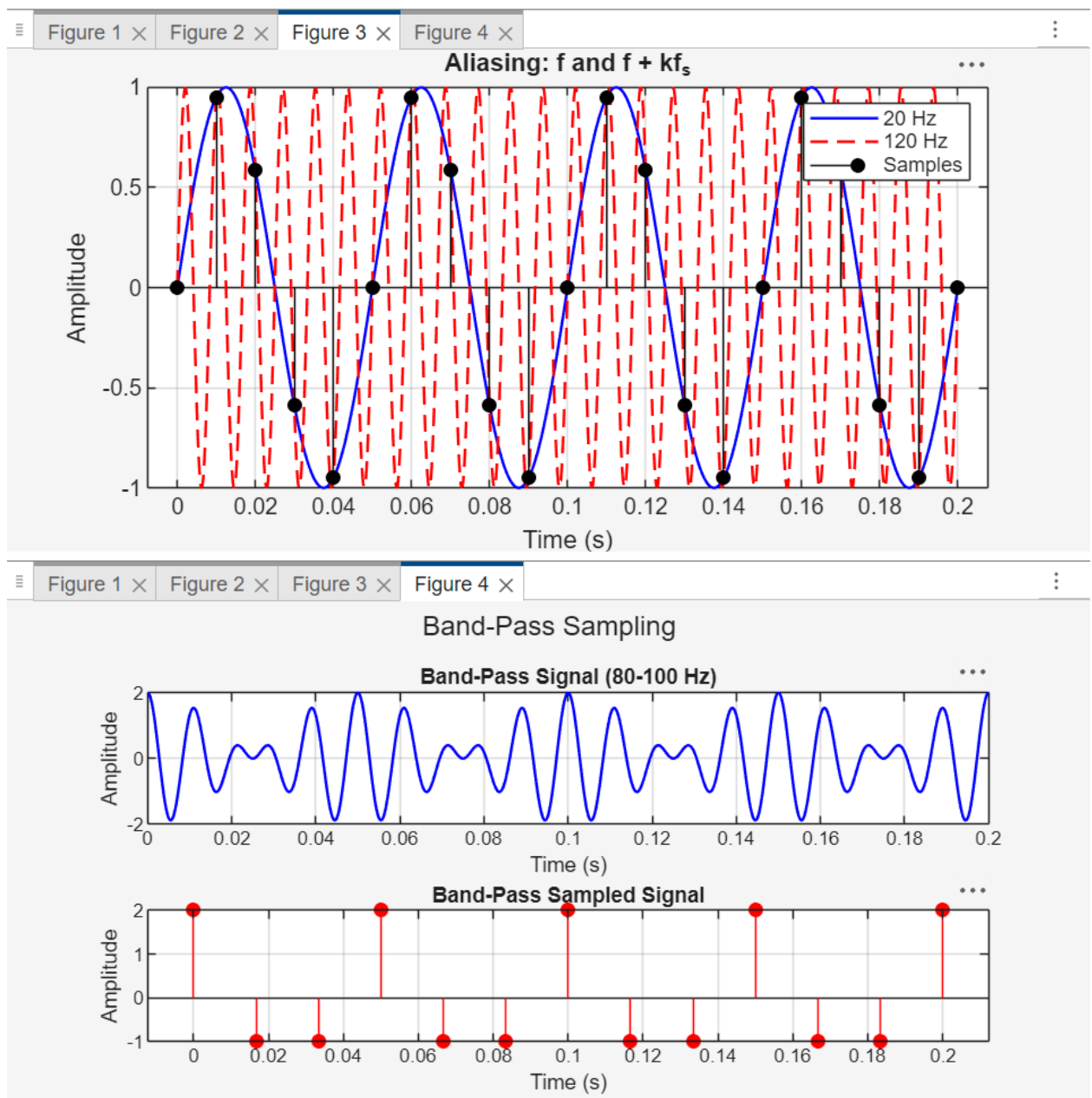
subplot(2,1,1);
plot(tb, xb, 'b', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Band-Pass Signal (80-100 Hz)');
subplot(2,1,2);
stem(tnb, xnb, 'r', 'filled');
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Band-Pass Sampled Signal');
sgtitle('Band-Pass Sampling');

```

Observations and Analysis:

Analyse spectrum of all signals





Observations:

1. Sampling of Continuous-Time Signal

A 10 Hz sinusoidal signal was sampled at a sampling frequency of 50 Hz. Since the sampling frequency is greater than twice the signal frequency, the Nyquist criterion is satisfied. The sampled signal correctly represents the original continuous-time signal.

2. High-Frequency Noise Aliasing

A 100 Hz high-frequency noise component was added to the 10 Hz signal and sampled at 50 Hz. The 100 Hz noise aliases to 0 Hz (DC), thereby affecting the sampled signal. This demonstrates the need for an anti-aliasing filter before sampling.

3. Aliasing: f and $f + kfs$

A 20 Hz signal and a 120 Hz signal were sampled at 100 Hz. Since:

$$120 = 20 + 1(100)$$

Both signals produced identical sample points. This demonstrates that higher-frequency signals can appear as lower-frequency signals after sampling.

4. Band-Pass Sampling

A band-pass signal containing frequency components between 80 Hz and 100 Hz was sampled at a lower sampling frequency. The experiment demonstrates the concept of band-pass sampling for band-limited signals.

Answer to the Question

(a) If you sample at 44 kHz, does the noise interfere with the signal?

- Yes, it does.
- Reasoning: The noise band (100 -- 102 kHz) folds back into the discrete spectrum via aliasing at $|f_{\text{noise}} - k \cdot f_s| = |100 - 102 - 2 \cdot 44| = 12 - 14 \text{ kHz}$. Because 12 -- 14 kHz falls directly inside the desired signal band (1 -- 20 kHz), it causes severe interference. An anti-aliasing low-pass filter must be applied before sampling to attenuate this noise.

(b) If you sample at 30 kHz, there is a problem. What is the problem?

- Under-sampling / Baseband Aliasing.
- Reasoning: The highest signal frequency is $f_{\text{max}} = 20 \text{ kHz}$, which requires a minimum Nyquist rate of $f_s \geq 40 \text{ kHz}$. Sampling at 30 kHz violates this condition. Signal components between 15 kHz and 20 kHz fold back onto 10 -- 15 kHz ($|30 - f|$), resulting in irreversible spectral distortion within the primary signal itself.

CONCLUSION

The experiment verified the sampling theorem and demonstrated the effects of aliasing and high-frequency noise. It was observed that insufficient sampling frequency causes spectral folding and distortion. The experiment also demonstrated the importance of anti-aliasing filtering and proper sampling of band-limited signals.

Signature of faculty in-charge